



Introduction

- The Ling Sound Test (Ling, 1976; 1989) is widely used internationally to verify access to key speech frequencies through hearing aids or cochlear implants
- Six English speech sounds (/m/, /u/, /a/, /i/, /f/, /s/) span the speech spectrum.
 - In Aotearoa New Zealand and Australia, it is routinely used in educational and clinical settings, often with the back vowel /ɔ/ added or in place of /u/ (Agung et al., 2005; McDonnell, 2014; Rees, 2019)
- These English Ling sounds do not align with the phonemic system of te reo Māori – the Indigenous language of Māori people of Aotearoa New Zealand
- Te reo Māori is undergoing active revitalisation, but existing hearing assessments in Aotearoa often require Māori-speaking children to respond to non-Māori phonology
 - The high-frequency phonemes /s/ and /f/ do not occur in te reo Māori, resulting in reduced linguistic and cultural validity for Māori tamariki (children)
- To address this gap, we developed **Te Whakamātautau Hoka**, a culturally responsive adaptation of the Ling test for use with Māori-speaking children. The test was co-designed with Indigenous communities, audiologists, speech-language therapists, and teachers through iterative consultation to ensure both cultural appropriateness and clinical practicality

Aims and Objectives

Aim:

- Develop a linguistically robust and culturally responsive te reo Māori adaptation of the Ling sound test.

Objectives

- Select Māori phonemes spanning the speech spectrum
- Co-design with professionals and Māori stakeholders
- Create calibrated mobile/web delivery
- Evaluate usability and acoustic characteristics

Development process

This research followed a He Awa Whiria / Braided Rivers approach (Macfarlane et al., 2015), braiding together indigenous Māori knowledge (mātauranga Māori) and Western knowledge traditions.

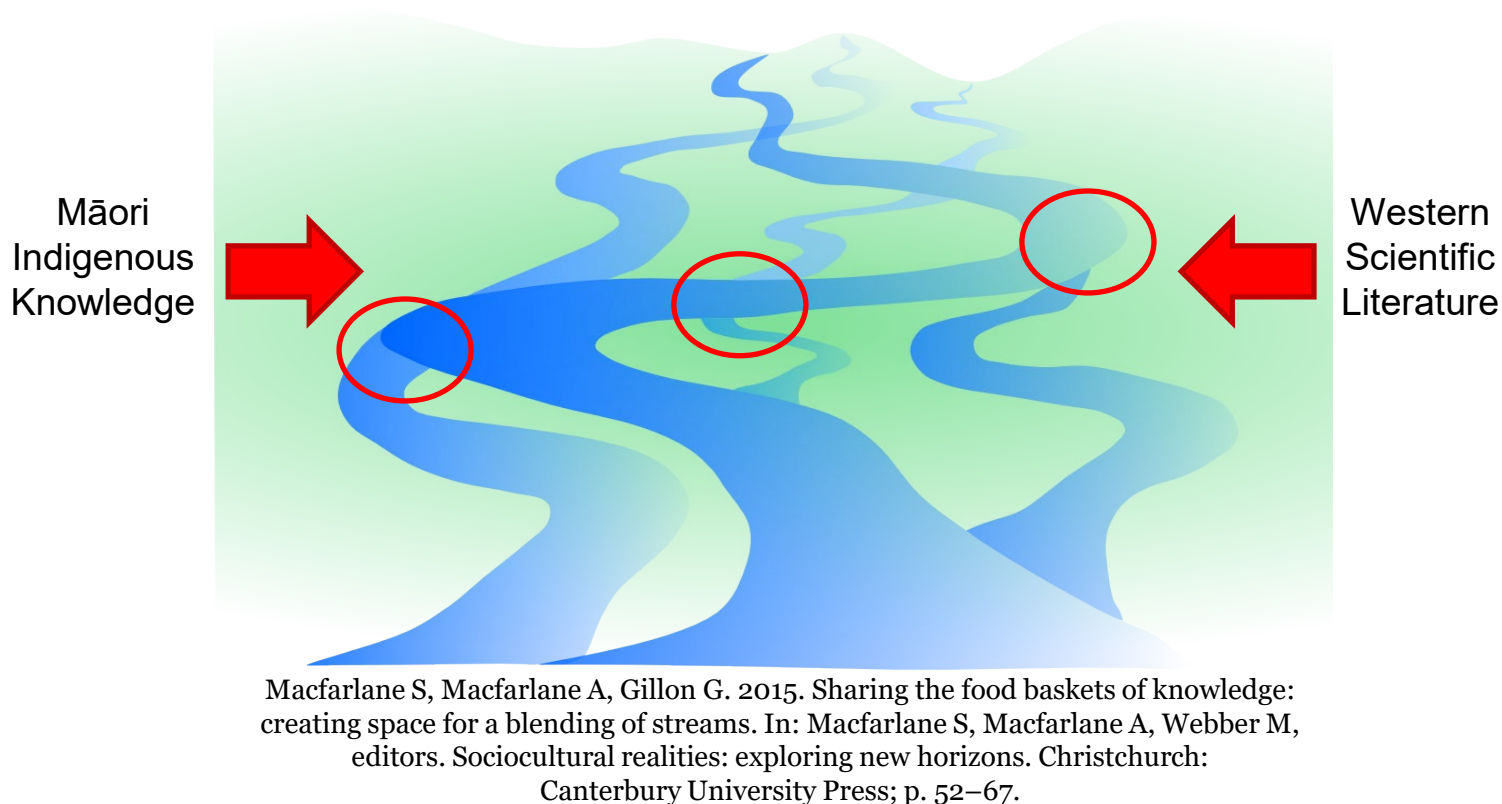
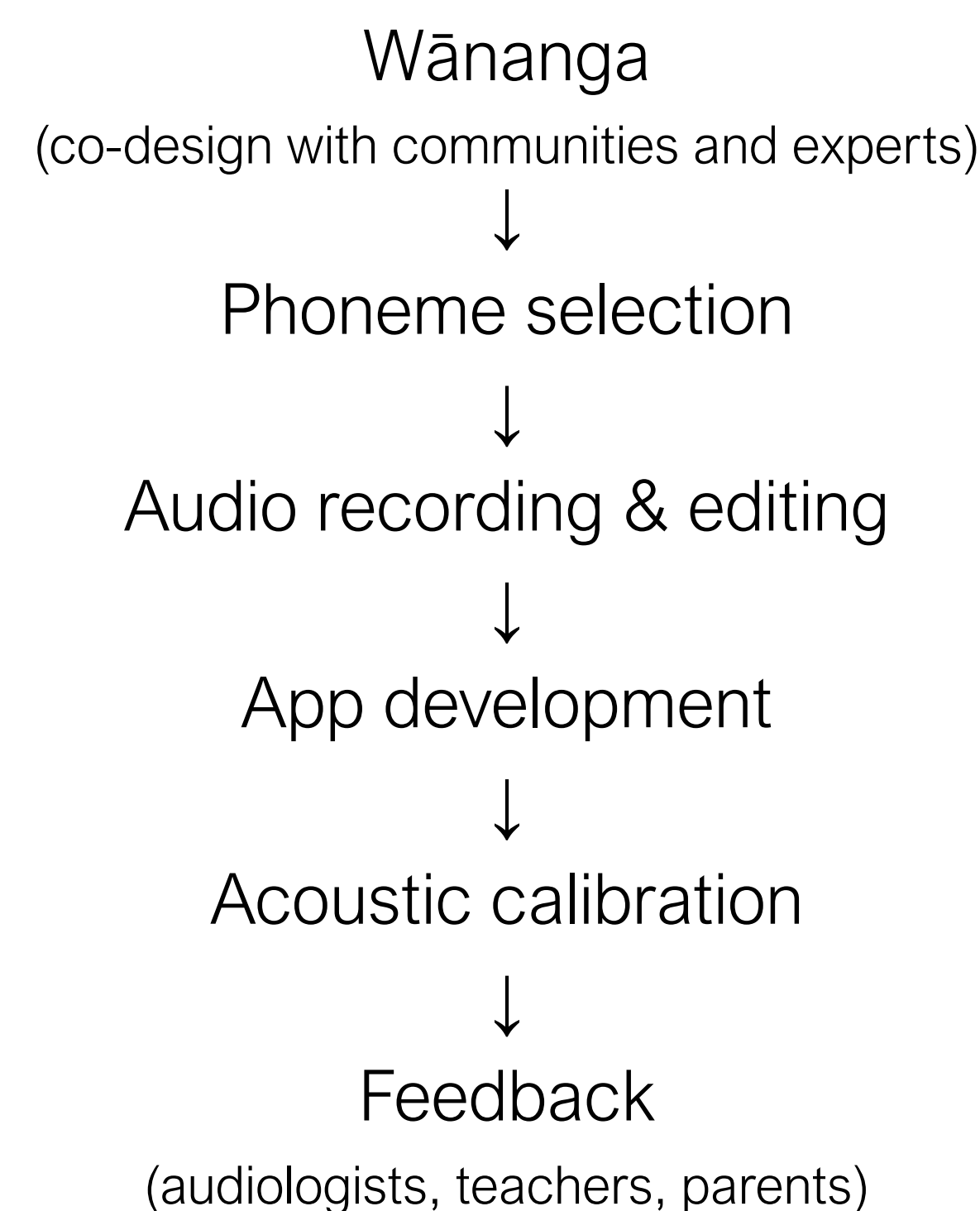


Figure 1: The He Awa Whiria collaboration model



Phoneme selection

Phonological analysis and acoustic measurements guided the selection of six representative Māori sounds for recording by a fluent speaker:

Low-frequency

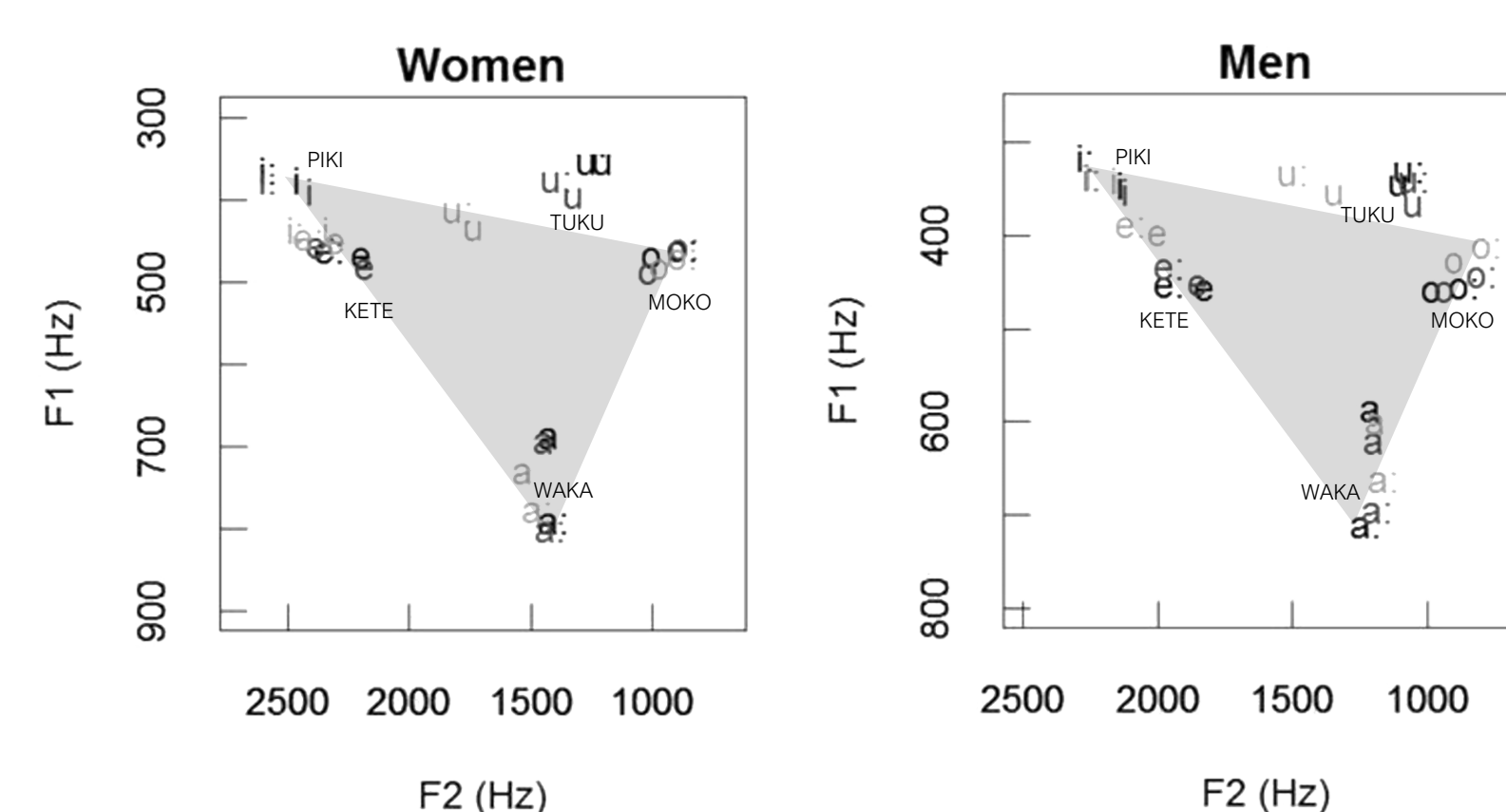
- /m/ – voiced bilabial nasal
- /o/ – MOKO – short mid-back rounded vowel
- /i/ – PIKI – short close-front unrounded vowel
- /a/ – WAKA – short open-central vowel

High-frequency

- /f/ – voiceless labiodental fricative
- /t/ – voiceless alveolar stop

Figure 2: Average F₁ and F₂ formant frequencies for male and female speakers (Watson et al., 2016).

- historical elders = black
- present-day elders = dark grey
- youth = light grey



- For comparison, stimuli were also recorded for TUKU (short close-back rounded vowel), KETE (short close-mid unrounded vowel).
- Variable pronunciation of the Māori “wh” digraph (commonly /f/, but also /w/ or /h/) prompted exploration of alternative high-frequency phonemes: /p/ (voiceless bilabial stop) and /h/ (voiceless glottal fricative).
- However, as demonstrated in Figure 3, only /f/ and /t/ had sufficient high-frequency acoustic energy to warrant inclusion in the finished stimulus set.
- The transient nature of /t/ required development of a repeated presentation delivery mode (e.g. x3, x5).

Acoustic analyses

Spectra of low-frequency sounds /m/, the vowels MOKO, PIKI, WAKA, KETE, and TUKU, and the consonants /h/, /p/, /t/, and /f/ are shown in Figure 3 below:

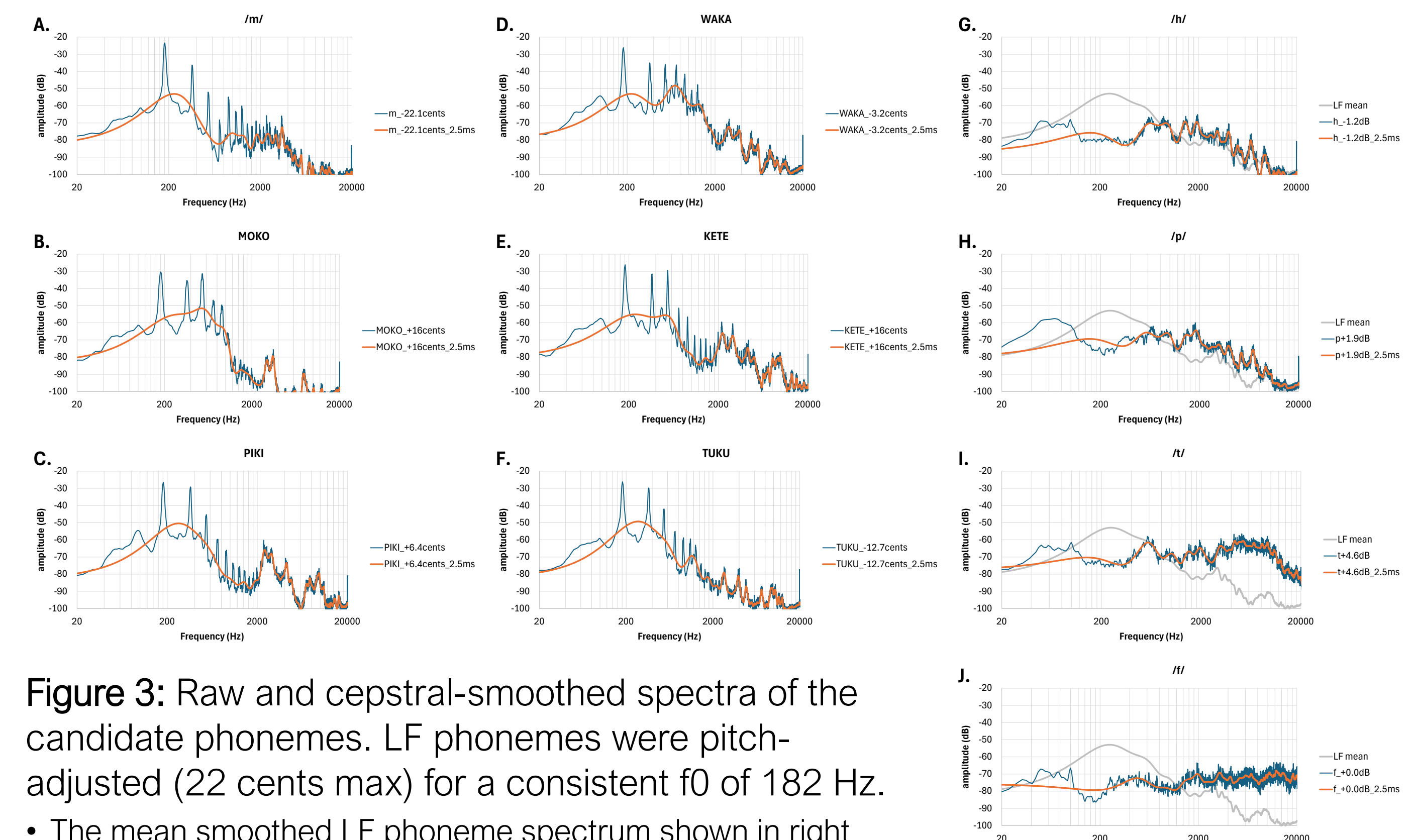


Figure 3: Raw and cepstral-smoothed spectra of the candidate phonemes. LF phonemes were pitch-adjusted (22 cents max) for a consistent f₀ of 182 Hz.

- The mean smoothed LF phoneme spectrum shown in right column for context.

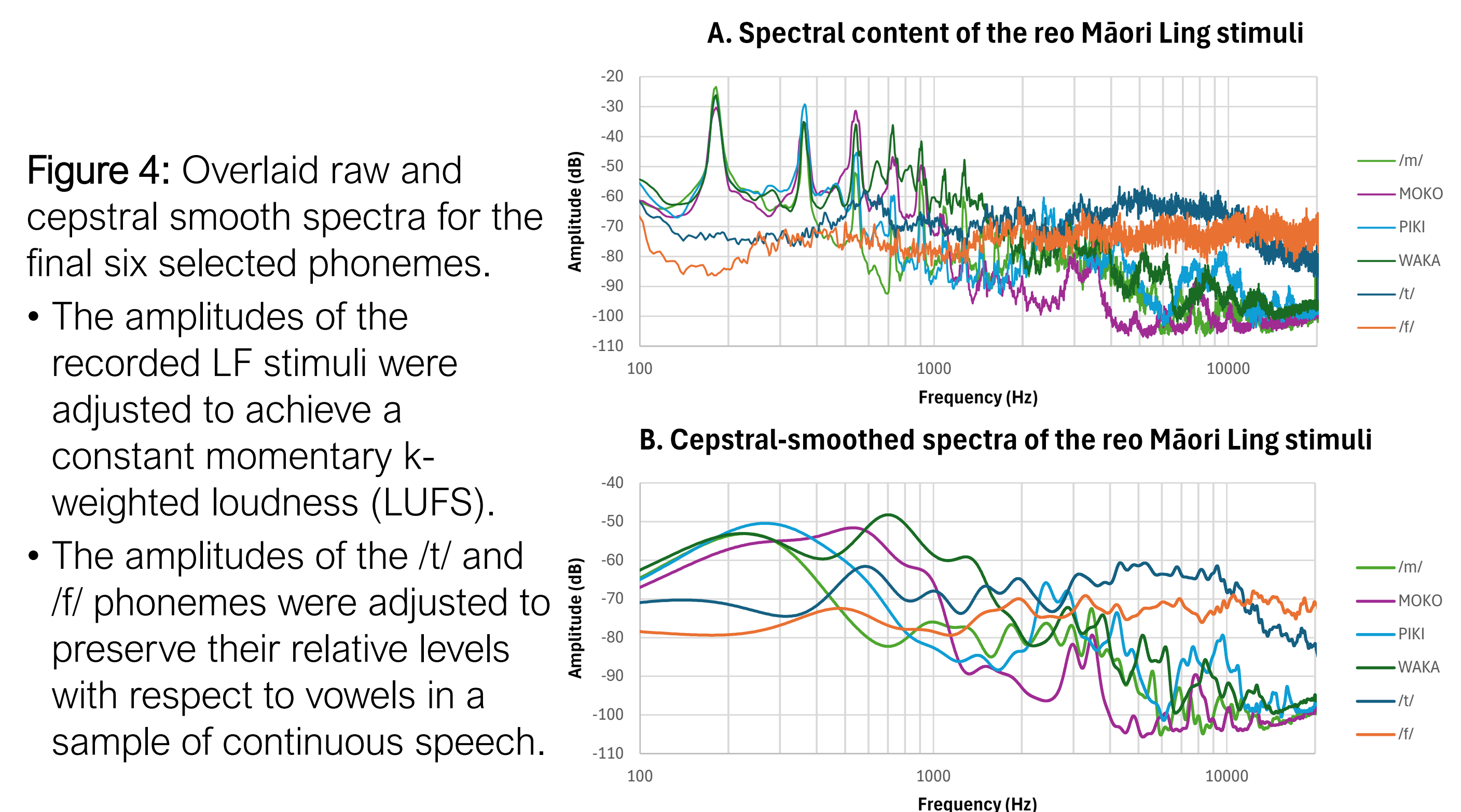


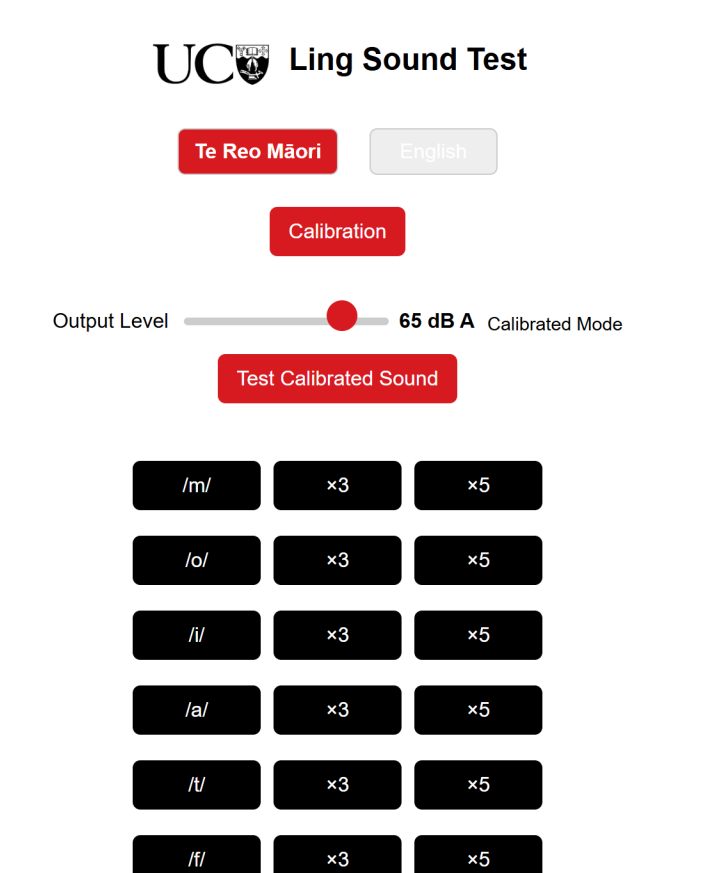
Figure 4: Overlaid raw and cepstral smooth spectra for the final six selected phonemes.

- The amplitudes of the recorded LF stimuli were adjusted to achieve a constant momentary k-weighted loudness (LUFS).
- The amplitudes of the /t/ and /f/ phonemes were adjusted to preserve their relative levels with respect to vowels in a sample of continuous speech.

App development

A responsive mobile/web application was developed to support consistent and accessible delivery of the reo Māori Ling sound test across educational and clinical settings.

- Responsive design for phones, tablets, and desktop devices
- Preloaded audio enabled low-latency offline playback
- Calibration tools supported dB A level adjustment using a sound-level meter or second device (e.g. iOS NIOSH SLM app)
- Te reo Māori and New Zealand English modes supports use across Aotearoa
- Early professional feedback indicates strong engagement and enthusiasm for a linguistically and culturally authentic tool



Conclusion

Te Whakamātautau Hoka – the reo Māori Ling sound test:

- Supports equitable and culturally meaningful hearing assessment in a revitalising Indigenous language.
- Demonstrates how linguistically and culturally grounded design can enhance engagement, accessibility, and clinical relevance in paediatric audiology.

References

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